University of Minho
School of Engineering

Intelligent Systems Associate Laboratory



Dados e Aprendizagem Automática

Ensemble Learning

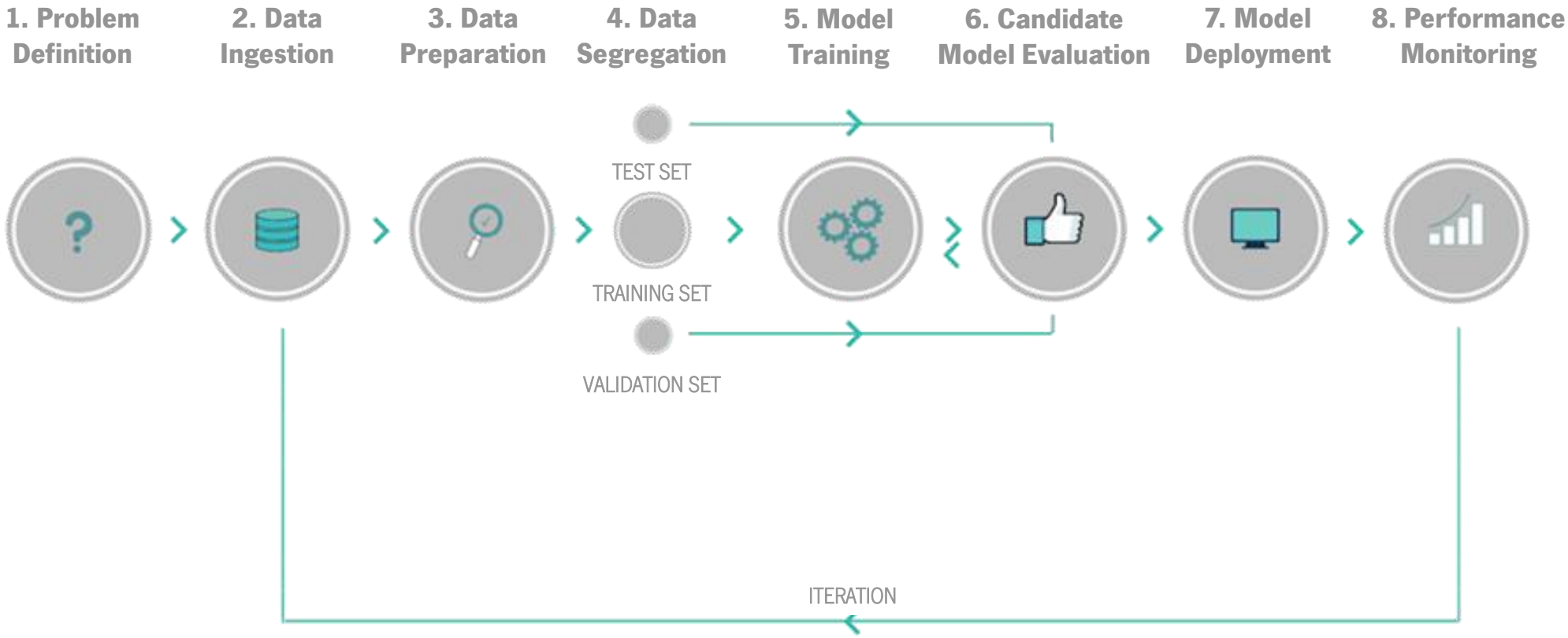
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Filipa Ferraz, Victor Alves, Dalila Durães, Francisco Marcondes, Guilherme Barbosa

Part VI

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Ensemble Learning

The Problem and the Data

Problem: development of a Machine Learning model that can **predict which passengers survived** the Titanic disaster.

Dataset: information regarding the **passengers** with 891 entries and 12 features, including:

- *PassengerId*
- *Survived*
- *Pclass*
- *Name*
- *Sex*
- *Age*
- *SibSp*
- *Parch*
- *Ticket*
- *Fare*
- *Cabin*
- *Embarked*

You may need to install `xgboost`. Use one of the following commands:

```
conda install -c conda-forge xgboost

pip install xgboost
```

EDA


```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

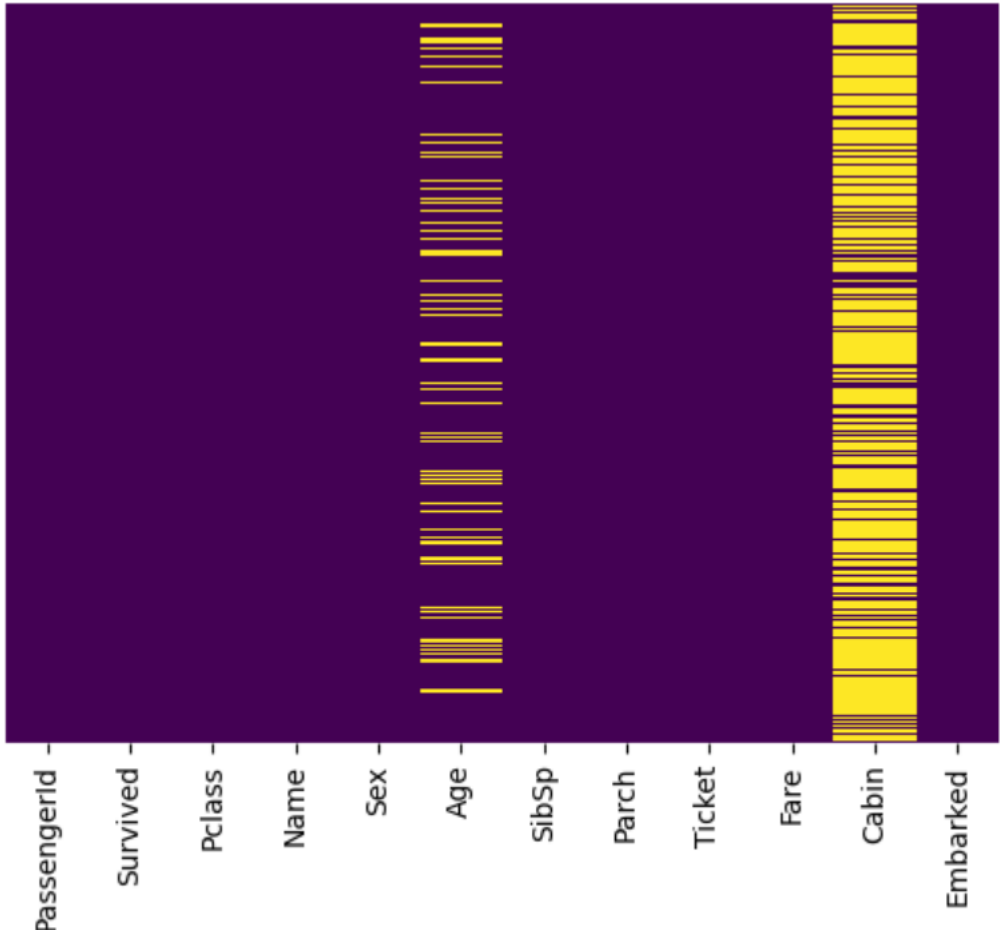
```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

EDA

We need to check the missing values:

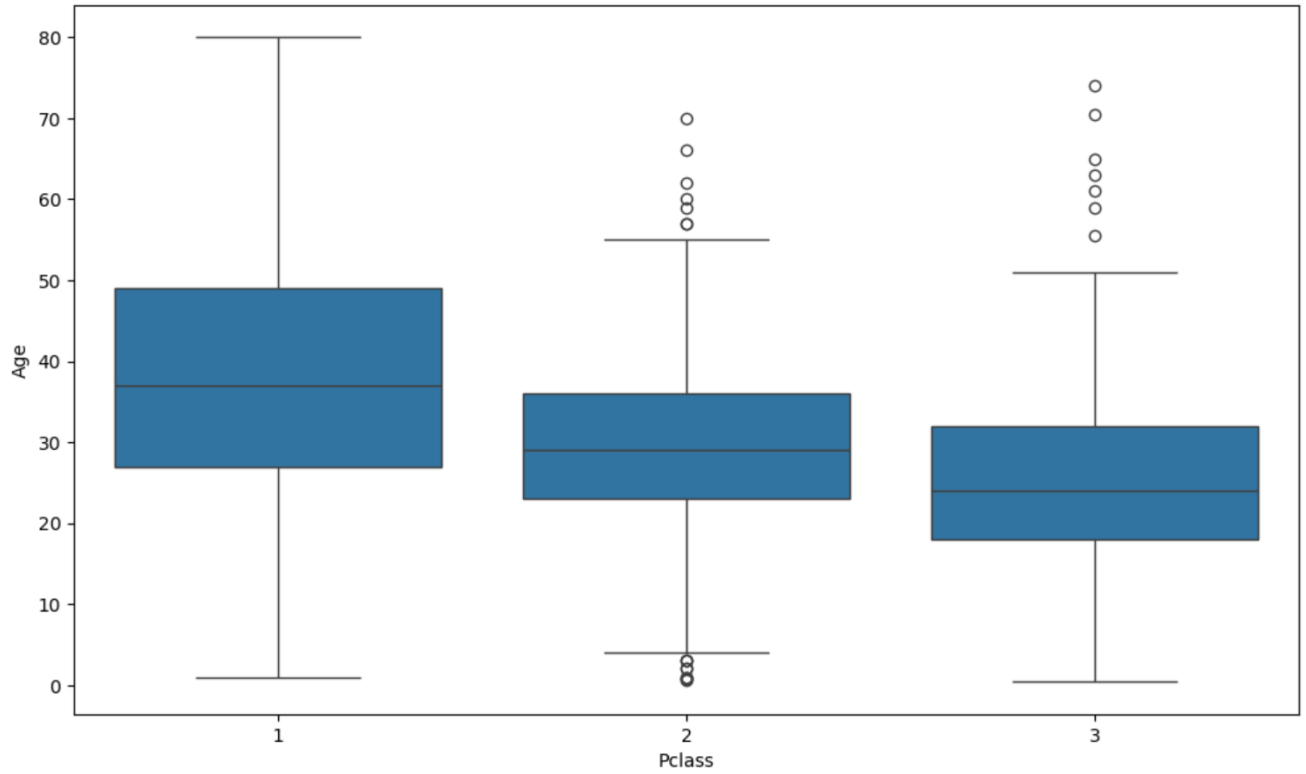
```
sns.heatmap(df.isnull(), yticklabels=False, cbar=False, cmap='viridis')
```



EDA/Data Preprocessing

Analyze the *Age* distribution by ticket class (*Pclass*):

```
plt.figure(figsize = (12, 7))
sns.boxplot(x = 'Pclass', y = 'Age', data = df)
```



We are going to impute missing *Age* values using the ticket class (*Pclass*):

```
df['Age'] = df['Age'].fillna(df.groupby('Pclass')['Age'].transform('median'))
```

Data Preprocessing

For now, we will remove the *Cabin*, *Sex*, *Embarked*, *Name* and *Ticket* features:

```
df.drop('Cabin', axis = 1, inplace = True)
df.dropna(inplace = True)
```

```
df.drop(['Sex', 'Embarked', 'Name', 'Ticket'], axis = 1,inplace = True)
```

```
df.head()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
0	1	0	3	22.0	1	0	7.2500
1	2	1	1	38.0	1	0	71.2833
2	3	1	3	26.0	0	0	7.9250
3	4	1	1	35.0	1	0	53.1000
4	5	0	3	35.0	0	0	8.0500

Define the X and y, and split the data:

```
X = df.drop('Survived', axis = 1)
y = df['Survived']
```

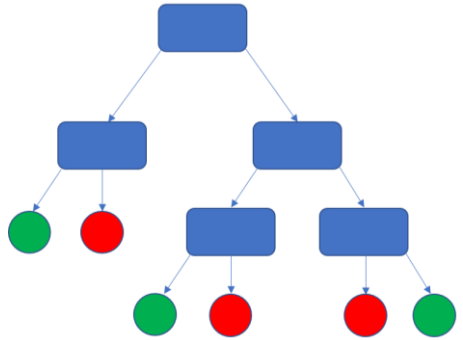
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.30, random_state = 2025)
```

```
X_train.shape, y_train.shape, X_test.shape, y_test.shape
```

```
((622, 6), (622,), (267, 6), (267,))
```

Comparing Models: Decision Tree

We are going try out some different models. We will implement Decision Tree, Logistic Regression and Random Forest models for comparison with Ensemble Learning models.



Decision Tree

```
dt_model = DecisionTreeClassifier(max_depth = 2, random_state = 2025)
```

```
dt_model.fit(X_train, y_train)
```

Evaluate the model and get the predictions:

```
dt_score = dt_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (dt_score * 100))
```

```
Accuracy: 68.16%
```

```
dt_predictions = dt_model.predict(X_test)
```

```
print(classification_report(y_test, dt_predictions))
```

	precision	recall	f1-score	support
0	0.72	0.78	0.75	162
1	0.61	0.53	0.57	105
accuracy			0.68	267
macro avg	0.66	0.66	0.66	267
weighted avg	0.68	0.68	0.68	267

Save the model's accuracy for later comparison:

```
results = {'DT': dt_score}
```

Comparing Models: Logistic Regression

Logistic Regression

```
lr_model = LogisticRegression(random_state=2025, solver='liblinear')
```

```
lr_model.fit(X_train, y_train)
```

```
lr_score = lr_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (lr_score * 100))
```

```
Accuracy: 67.79%
```

```
lr_predictions = lr_model.predict(X_test)
```

```
print(classification_report(y_test, lr_predictions))
```

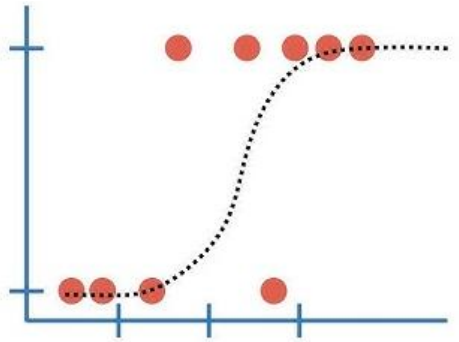
```
              precision    recall  f1-score   support

           0       0.68      0.89      0.77       162
           1       0.67      0.35      0.46       105

    accuracy                           0.68       267
   macro avg       0.68      0.62      0.62       267
weighted avg       0.68      0.68      0.65       267
```

```
results['LogR'] = lr_score
```

```
sklearn.linear_model.LogisticRegression(penalty='l2', *, dual=False, tol=0.0001,
C=1.0, fit_intercept=True, intercept_scaling=1, class_weight=None,
random_state=None, solver='lbfgs', max_iter=100,
multi_class='deprecated', verbose=0, warm_start=False, n_jobs=None,
l1_ratio=None)
```



Comparing Models: Random Forest

Random Forest

```
rf_model = RandomForestClassifier()
```

```
rf_model.fit(X_train, y_train)
```

```
rf_score = rf_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (rf_score * 100))
```

```
Accuracy: 71.16%
```

```
rf_predictions = rf_model.predict(X_test)
```

```
print(classification_report(y_test, rf_predictions))
```

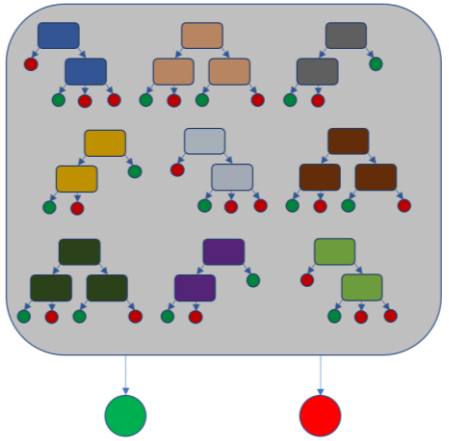
```
              precision    recall  f1-score   support

           0       0.73      0.83      0.78       162
           1       0.67      0.53      0.59       105

    accuracy                           0.71       267
   macro avg       0.70      0.68      0.68       267
weighted avg       0.71      0.71      0.70       267
```

```
results['RF'] = rf_score
```

```
sklearn.ensemble.RandomForestClassifier(n_estimators=100, *, criterion='gini',
                max_depth=None, min_samples_split=2, min_samples_leaf=1,
                min_weight_fraction_leaf=0.0, max_features='sqrt',
                max_leaf_nodes=None, min_impurity_decrease=0.0, bootstrap=True,
                oob_score=False, n_jobs=None, random_state=None, verbose=0,
                warm_start=False, class_weight=None, ccp_alpha=0.0,
                max_samples=None, monotonic_cst=None)
```

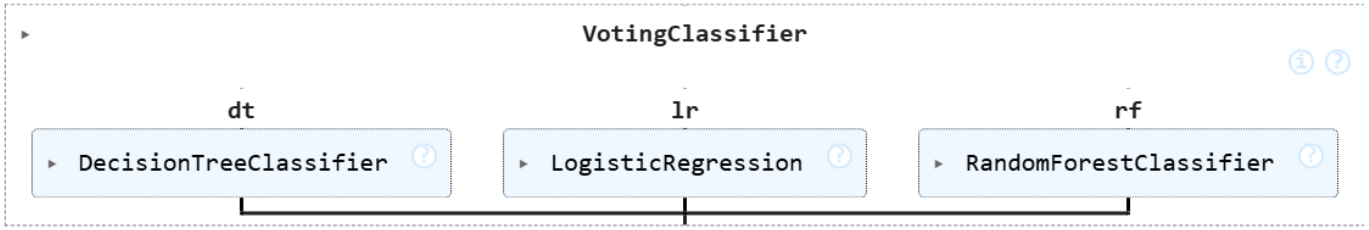


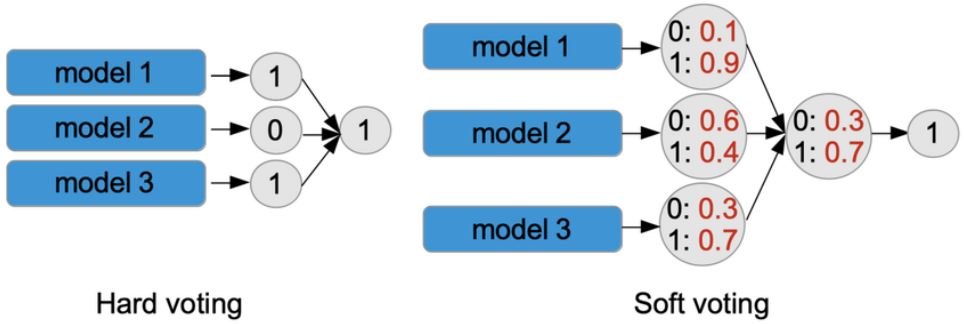
Ensemble Learning Models: Max Voting

Majority Class Labels (Majority/Hard Voting)

```
hvt_model = VotingClassifier(estimators = [("dt", dt_model), ("lr", lr_model), ("rf", rf_model)],
                             voting = 'hard', weights = [2, 1, 2])
```

```
hvt_model.fit(X_train, y_train)
```






```
sklearn.ensemble.VotingClassifier(estimators,
                *, voting='hard', weights=None,
                n_jobs=None,
                flatten_transform=True,
                verbose=False)
```

Max voting involves fitting several models, each making a prediction and voting on each sample. Only the class with the most votes is included in the final prediction.

Ensemble Learning Models: Max Voting


```
for model, label in zip([dt_model, lr_model, rf_model, hvt_model], ['dt', 'lr', 'rf', 'Ensemble']):
    hvt_score = cross_val_score(model, X_test, y_test, scoring = 'accuracy', cv = 5)
    print("Accuracy: %0.2f (+/- %0.2f) [%s]" % (hvt_score.mean(), hvt_score.std(), label))
```

```
Accuracy: 0.60 (+/- 0.04) [dt]
Accuracy: 0.66 (+/- 0.07) [lr]
Accuracy: 0.64 (+/- 0.06) [rf]
Accuracy: 0.65 (+/- 0.08) [Ensemble]
```

```
hvt_score = hvt_model.score(X_test, y_test)
print("Accuracy: %.2f%%" % (hvt_score*100))
```

```
Accuracy: 69.66%
```

```
hvt_predictions = hvt_model.predict(X_test)
```

```
print(classification_report(y_test, hvt_predictions))
```

```
              precision    recall  f1-score   support

           0       0.71      0.84      0.77       162
           1       0.66      0.48      0.55       105

    accuracy                           0.70       267
   macro avg       0.68      0.66      0.66       267
weighted avg       0.69      0.70      0.68       267
```

```
results['HVotW']= hvt_score
```

Ensemble Learning Models: BAGGing *(Bootstrap AGGregating)*

We will use the **StratifiedShuffleSplit** technique which is another form of splitting.

```
sss = StratifiedShuffleSplit(n_splits = 10, test_size = 20, random_state = 2025)
```

```
sklearn.model_selection.StratifiedShuffleSplit(n_splits=10, *,
                test_size=None, train_size=None, random_state=None)
```

Implement the **Bagging Classifier** with Decision Tree:

```
bg_dt_model = BaggingClassifier(estimator = dt_model, bootstrap = True)
```

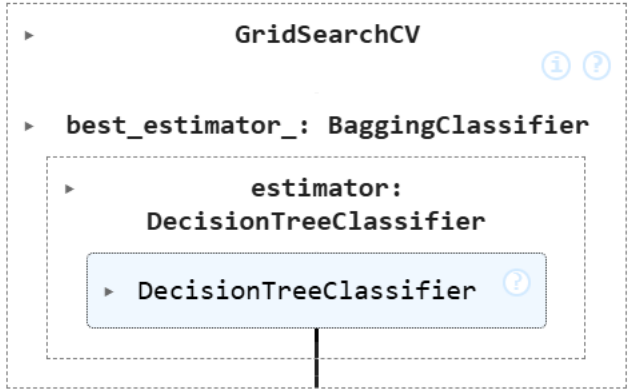
```
n_estimators = [10, 40, 60, 80, 100, 160]
```

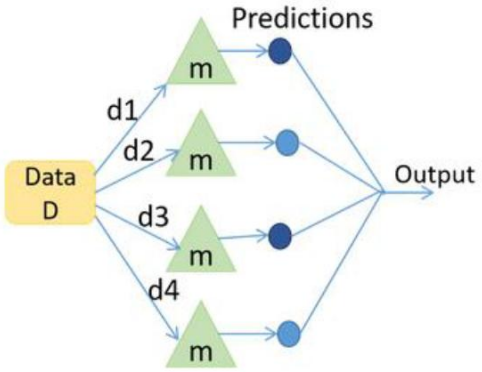
```
parameters = {'n_estimators': n_estimators}
```

```
grid_bg = GridSearchCV(estimator = bg_dt_model, param_grid = parameters, cv = sss)
```

```
grid_bg.fit(X_train, y_train)
```

```
sklearn.ensemble.BaggingClassifier(estimator=None,
                n_estimators=10, *, max_samples=1.0,
                max_features=1.0, bootstrap=True,
                bootstrap_features=False, oob_score=False,
                warm_start=False, n_jobs=None, random_state=None,
                verbose=0)
```





BAGGing involves creating different decision trees from the same dataset and evaluating them by calculating the average of their predictions.

Ensemble Learning Models: Bagging

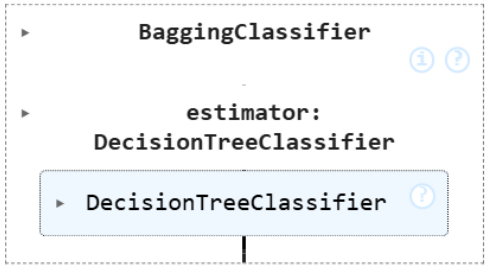
Best estimator

```
bst_bg_model = grid_bg.best_estimator_
```

```
print(bst_bg_model)
```

```
BaggingClassifier(estimator=DecisionTreeClassifier(max_depth=2,
                                                   random_state=2025),
                  n_estimators=80)
```

```
bst_bg_model.fit(X_train, y_train)
```




```
bst_bg_score = bst_bg_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (bst_bg_score*100))
```

```
Accuracy: 69.29%
```

```
bg_predictions = bst_bg_model.predict(X_test)
```

```
print(classification_report(y_test, bg_predictions))
```

```
              precision    recall  f1-score   support

           0       0.69      0.88      0.78       162
           1       0.69      0.40      0.51       105

    accuracy                           0.69       267
   macro avg       0.69      0.64      0.64       267
weighted avg       0.69      0.69      0.67       267
```

```
results['BaggDT']= bst_bg_score
```

Ensemble Learning Models: Bagging

Bagging with **Random Forest**:

```
bg_rf_model = RandomForestClassifier(bootstrap = False, max_depth = 2, verbose = 1)
```

```
bg_rf_model.fit(X_train, y_train)
```

```
bg_rf_score = bg_rf_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (rf_score * 100))
```

```
Accuracy: 71.16%
```

```
bg_rf_predictions = bg_rf_model.predict(X_test)
```

```
print(classification_report(y_test, bg_rf_predictions))
```

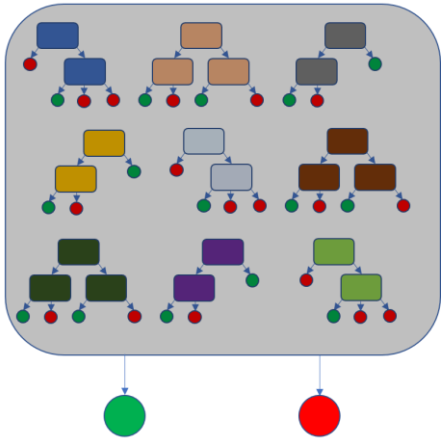
```
              precision    recall  f1-score   support

           0       0.71      0.81      0.76       162
           1       0.62      0.49      0.55       105

    accuracy                           0.68       267
   macro avg       0.67      0.65      0.65       267
weighted avg       0.67      0.68      0.67       267
```

```
results['BaggRF'] = bg_rf_score
```

```
sklearn.ensemble.RandomForestClassifier(n_estimators=100, *,
    criterion='gini', max_depth=None, min_samples_split=2,
    min_samples_leaf=1, min_weight_fraction_leaf=0.0,
    max_features='sqrt', max_leaf_nodes=None,
    min_impurity_decrease=0.0, bootstrap=True, oob_score=False,
    n_jobs=None, random_state=None, verbose=0, warm_start=False,
    class_weight=None, ccp_alpha=0.0, max_samples=None,
    monotonic_cst=None)
```



Ensemble Learning Models: Boosting

Gradient Boosting

```
gbc_model = GradientBoostingClassifier(n_estimators = 100, learning_rate = 1.0,
                                       max_depth = 1, random_state = 2025)
```

```
gbc_model.fit(X_train, y_train)
```

```
gbc_score = gbc_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (gbc_score*100))
```

```
Accuracy: 68.54%
```

```
gbc_predictions = gbc_model.predict(X_test)
```

```
print(classification_report(y_test, gbc_predictions))
```

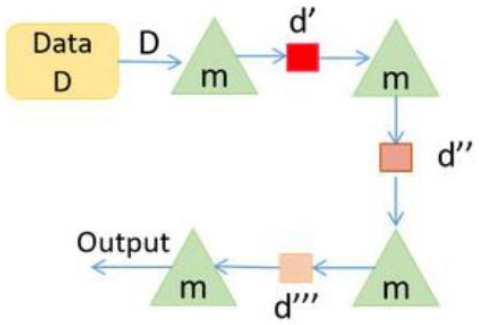
```
              precision    recall  f1-score   support

           0       0.71      0.80      0.76       162
           1       0.62      0.50      0.56       105

    accuracy                           0.69       267
   macro avg       0.67      0.65      0.66       267
weighted avg       0.68      0.69      0.68       267
```

```
results['GB'] = gbc_score
```

```
sklearn.ensemble.GradientBoostingClassifier(*, loss='log_loss',
                learning_rate=0.1, n_estimators=100, subsample=1.0,
                criterion='friedman_mse', min_samples_split=2,
                min_samples_leaf=1, min_weight_fraction_leaf=0.0,
                max_depth=3, min_impurity_decrease=0.0, init=None,
                random_state=None, max_features=None, verbose=0,
                max_leaf_nodes=None, warm_start=False,
                validation_fraction=0.1, n_iter_no_change=None,
                tol=0.0001, ccp_alpha=0.0)
```



Boosting involves adding ensemble members sequentially to correct the predictions made by previous models, producing a weighted average of the predictions.

Ensemble Learning Models: Boosting

XGBoost

```
xgb_model = XGBClassifier(max_depth = 1, objective = 'reg:squarederror')
```

```
xgb_model.fit(X_train, y_train)
```

```
xgb_score = xgb_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (xgb_score * 100))
```

```
Accuracy: 67.42%
```

```
xgb_predictions = xgb_model.predict(X_test)
```

```
print(classification_report(y_test, xgb_predictions))
```

```
              precision    recall  f1-score   support

           0       0.69      0.84      0.76       162
           1       0.63      0.42      0.50       105

    accuracy                           0.67       267
   macro avg       0.66      0.63      0.63       267
weighted avg       0.67      0.67      0.66       267
```

```
results['XGB'] = xgb_score
```

```
xgboost.sklearn.XGBClassifier(base_score=None, booster=None,
              callbacks=None, colsample_bylevel=None,
              colsample_bynode=None, colsample_bytree=None,
              device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None,
              feature_types=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None,
              learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None,
              missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None,
              n_jobs=None, num_parallel_tree=None,
              random_state=None, ...)
```

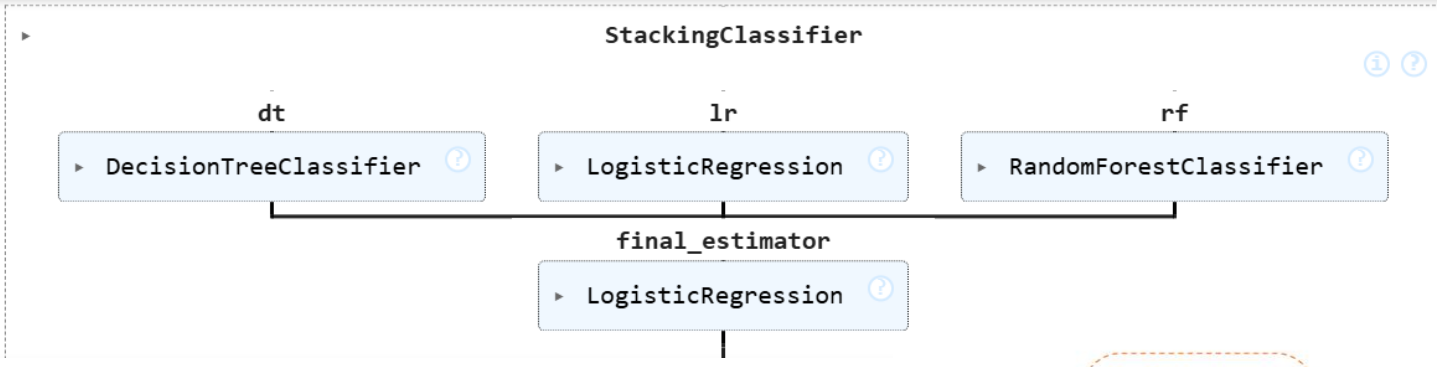
Ensemble Learning Models: Stacking

Stacking Classifier

```
estimators = [("dt", dt_model), ("lr", lr_model), ("rf", rf_model)]
```

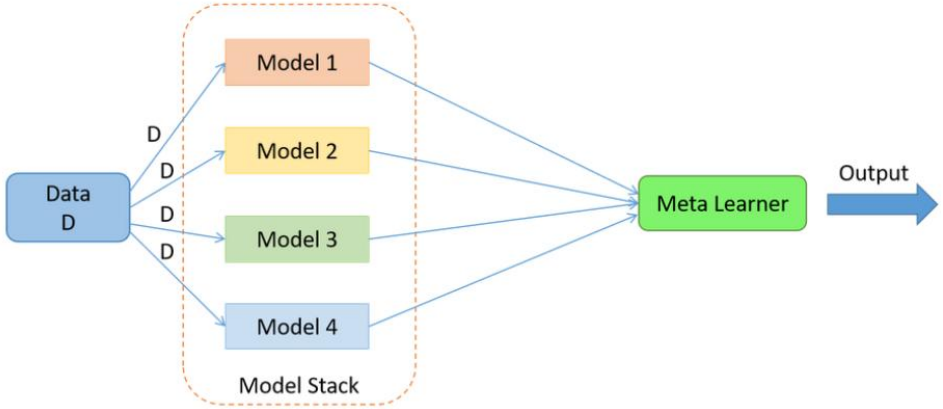
```
st_model = StackingClassifier(estimators = estimators,
                              final_estimator = LogisticRegression())
```

```
st_model.fit(X_train, y_train)
```




```
sklearn.ensemble.StackingClassifier(estimators,
                                    final_estimator=None, *, cv=None,
                                    stack_method='auto', n_jobs=None,
                                    passthrough=False, verbose=0)
```

Stacking involves fitting different types of model to the same dataset and using one model to learn the most effective way to combine the predictions.



Ensemble Learning Models: Stacking


```
st_score = st_model.score(X_test, y_test)
```

```
print("Accuracy: %.2f%%" % (st_score*100))
```

```
Accuracy: 67.79%
```

```
st_predictions = st_model.predict(X_test)
```

```
print(classification_report(y_test, st_predictions))
```

```
              precision    recall  f1-score   support

           0       0.70      0.83      0.76       162
           1       0.63      0.44      0.52       105

    accuracy                           0.68       267
   macro avg       0.66      0.64      0.64       267
weighted avg       0.67      0.68      0.66       267
```

```
results['Stack']= st_score
```

Model Accuracy Comparison

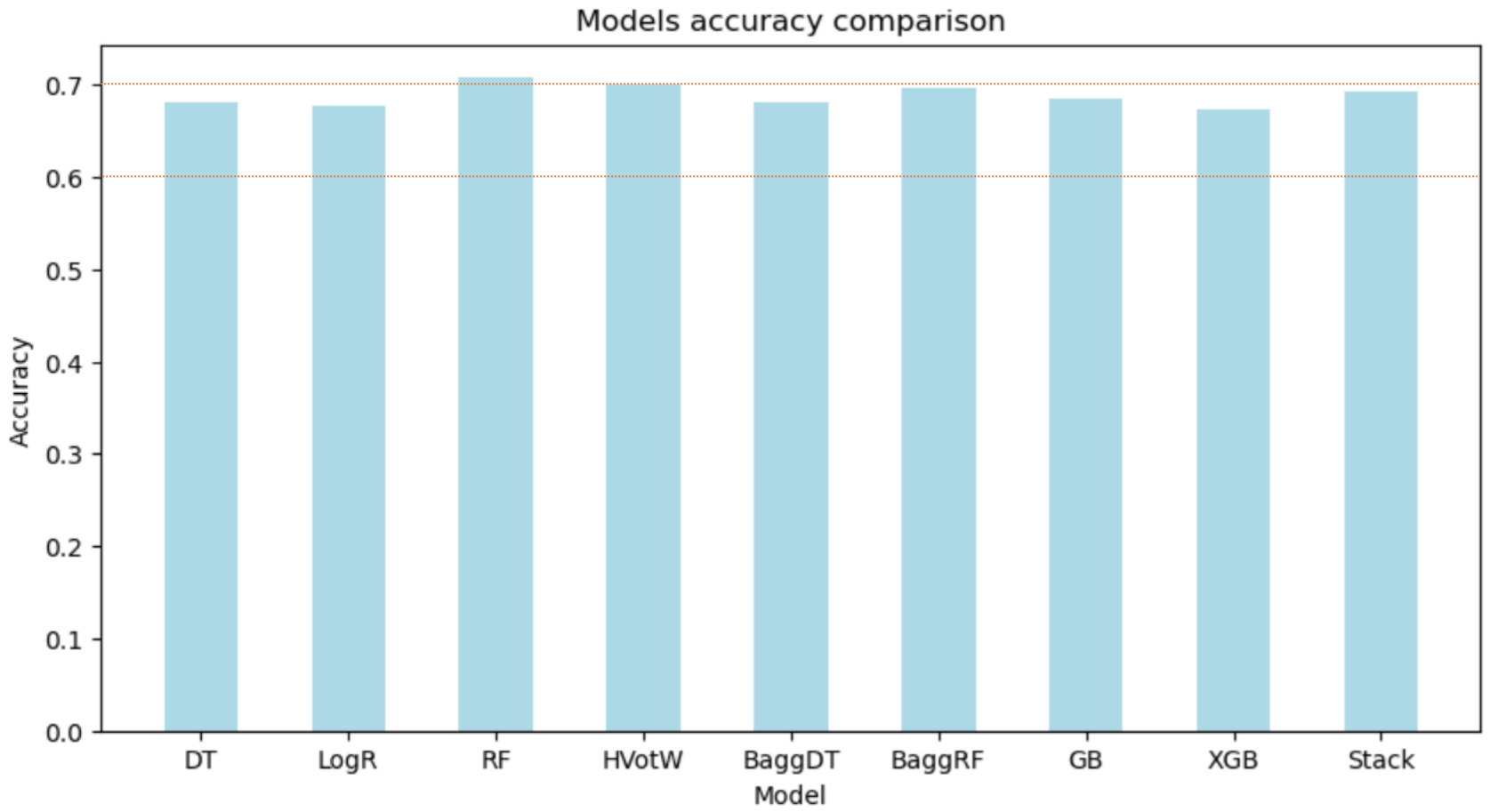
Create a bar chart showing the accuracy of each model:

```
fig = plt.figure(figsize = (10, 5))

mod = list(results.keys())
acc =list(results.values())

plt.bar(mod, acc, color ='lightblue',
        width = 0.5)

plt.xlabel("Model")
plt.ylabel("Accuracy")
plt.title("Models accuracy comparison")
plt.show()
```



Model Accuracy Comparison

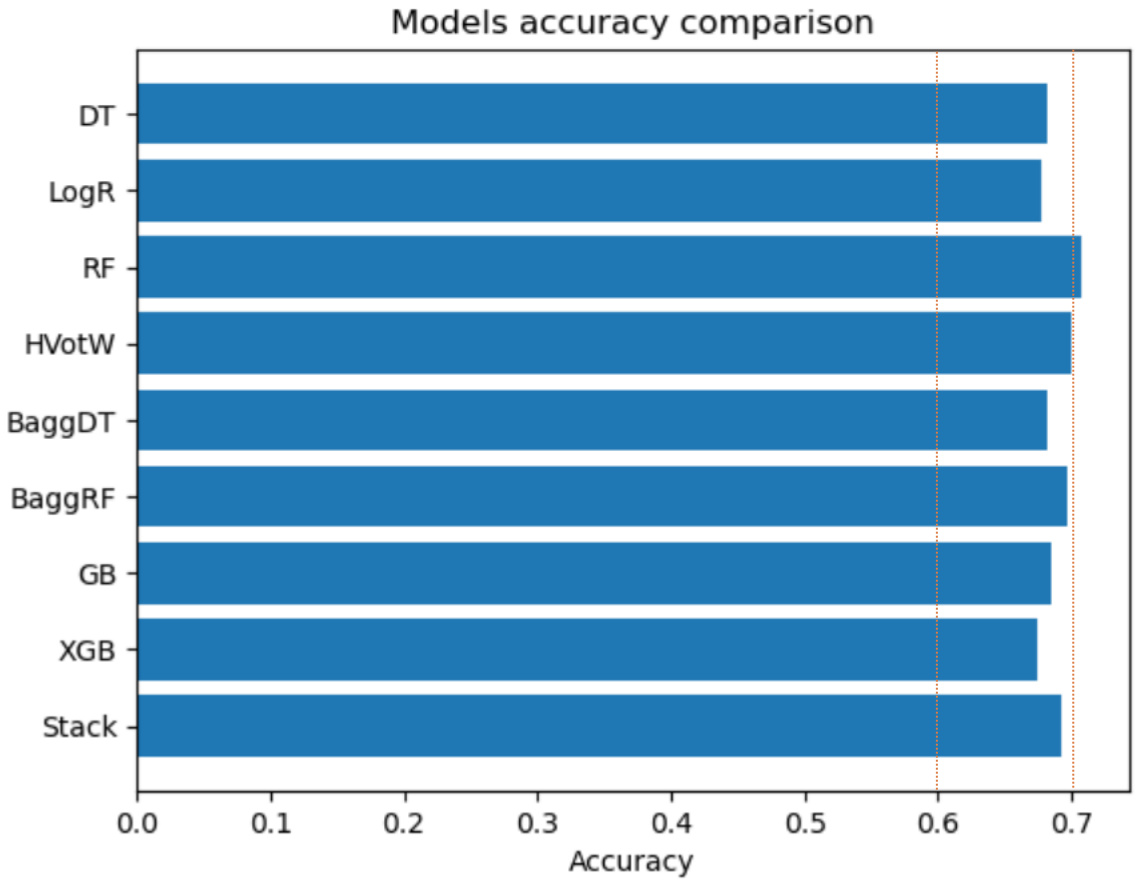
We can visualize this using horizontal bars:

```
fig, ax = plt.subplots()

y_values = np.arange(len(mod))

ax.barh(y_values, acc, align='center')
ax.set_yticks(y_values, labels = mod)
ax.invert_yaxis()
ax.set_xlabel('Accuracy')
ax.set_title('Models accuracy comparison')

plt.show()
```



Model Accuracy Comparison

Alternatively, we can print the results:

```
print("Models accuracy comparison")
for key, value in results.items():
    print("%s \t %.2f" % (key, value))
```

```
Models accuracy comparison
DT       0.68
LogR     0.68
RF       0.71
HVotW    0.70
BaggDT   0.68
BaggRF   0.70
GB       0.69
XGB      0.67
Stack    0.69
```

Which model performed better?
Which models are worth tuning?

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Which model performed better?
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Hands On